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EX NO: 5

ASSIGNMENT -5

Form-Servlet Communication

AIM

To design and develop a web-based authentication and registration system using **HTML, CSS, JavaScript, and Java Servlets**, demonstrating form-servlet communication, server-side validation, and page redirection based on user credentials.

DESCRIPTIONS

<web-app> — Root tag that defines and configures the web application.

<servlet> — Declares a servlet and its configuration details.

<servlet-name> — Specifies the logical name of the servlet.

<servlet-class> — Defines the Java class that implements the servlet.

<servlet-mapping> — Maps the servlet to a client request URL.

<url-pattern> — Specifies the URL path used to access the servlet.

HttpServlet — Base class extended to create an HTTP servlet.

doPost() — Handles client requests sent using the POST method.

HttpServletRequest — Object used to receive data from the client request.

HttpServletResponse — Object used to send response back to the client.

request.getParameter() — Retrieves form input values from the request.

response.sendRedirect() — Redirects the client to another web page.

HttpSession — Creates or accesses session data for a user.

request.getSession() — Returns the current session object.

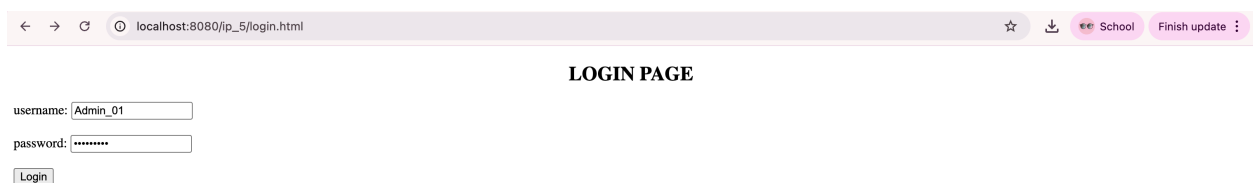
session.setAttribute() — Stores data in the session scope.

response.setContentType() — Sets the MIME type of the response sent to the browser.

PrintWriter out = response.getWriter() — Sends text/HTML output to the client.

RESULT

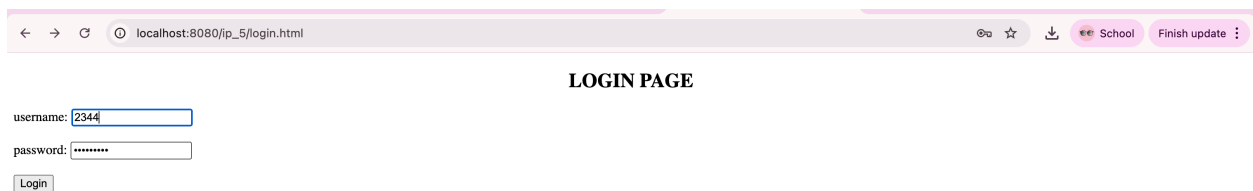
The web-based authentication application was successfully designed and developed using HTML and Java Servlets. Form–Servlet communication was established, server-side validation was implemented for username and password, and user authentication was performed. Based on the entered credentials, the system successfully redirected users to the appropriate success or error page.



A screenshot of a web browser showing the login page. The address bar displays 'localhost:8080/ip_5/login.html'. The page title is 'LOGIN PAGE'. It features a form with 'username:' and 'password:' labels, each followed by a text input field. The username field contains 'Admin_01' and the password field contains seven asterisks. A 'Login' button is positioned below the password field. The browser's top bar includes navigation icons, a star icon, a download icon, and two buttons labeled 'School' and 'Finish update'.



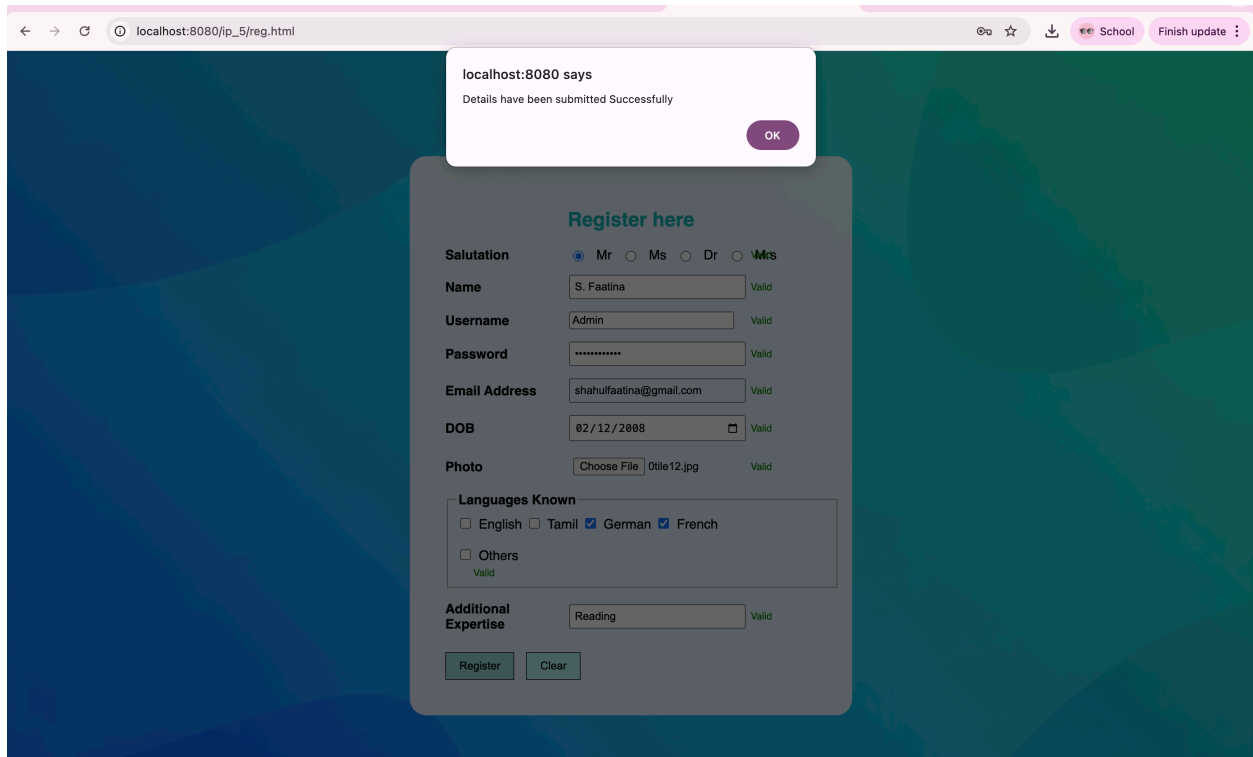
A screenshot of a web browser showing the 'Login Successful' page. The address bar displays 'localhost:8080/ip_5/success.html'. The page title is 'Login Successful'. The main content area displays 'Welcome! You have logged in successfully.' The browser's top bar includes navigation icons, a star icon, a download icon, and two buttons labeled 'School' and 'Finish update'.



A screenshot of a web browser showing the login page. The address bar displays 'localhost:8080/ip_5/login.html'. The page title is 'LOGIN PAGE'. It features a form with 'username:' and 'password:' labels, each followed by a text input field. The username field contains '234' and the password field contains seven asterisks. A 'Login' button is positioned below the password field. The browser's top bar includes navigation icons, a star icon, a download icon, and two buttons labeled 'School' and 'Finish update'.



A screenshot of a web browser showing the 'Invalid Username or Password' page. The address bar displays 'localhost:8080/ip_5/error.html'. The page title is 'Invalid Username or Password'. The main content area displays 'Please try again.' followed by a blue hyperlink labeled 'Go Back to Login Page'. The browser's top bar includes navigation icons, a star icon, a download icon, and two buttons labeled 'School' and 'Finish update'.



LEARNING OUTCOME

1. Understood the communication between HTML forms and Java Servlets using GET and POST methods.
2. Implemented server-side validation for username and password based on the specified rules.
3. Developed authentication logic to verify user credentials using predefined values.
4. Applied page navigation using `response.sendRedirect()` for success and failure scenarios.
5. Gained knowledge of session handling to store and display error messages.
6. Learned the deployment and execution of servlet-based web applications on the Apache Tomcat server.